

Practice Question

TID	List of items	Given the transaction data, find all frequent item sets having minimum support count = 2
T100	I1, I2, I5	
T200	I2, I4	
T300	I2, I3	
T400	I1, I2, I4	
T500	I1, I3	
T600	I2, I3	
T700	I1, I3	
T800	I1, I2, I3, I5	
T900	I1, I2, I3	

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C1:

I1. 6

I2. 7

I3. 6

I4. 2

I5. 2

L1:

I1

I2

I3

I4

I5

C2:

{I1, I2}. 4

{I1, I3}. 4

$\{i1, i4\}$ 1 <- eliminate
 $\{i1, i5\}$. 2
 $\{i2, i3\}$. 4
 $\{i2, i4\}$ 2
 $\{i2, i5\}$. 2
 $\{i3, i4\}$ 0 <-- eliminate
 $\{i3, i5\}$ 1 $\leftarrow$ eliminate

L2:

$\{i1, i2\}$.
 $\{i1, i3\}$.
 $\{i1, i5\}$.
 $\{i2, i3\}$.
 $\{i2, i4\}$
 $\{i2, i5\}$.

C3:

$\{i1, i2, i3\}$ 2
 $\{i1, i2, i5\}$. 2
 $\{i1, i3, i5\}$. $\{i3, i5\}$ is infrequent -> prune
 $\{i2, i3, i4\}$ $\{i3, i4\}$ is infrequent $\rightarrow$ prune
 $\{i2, i3, i5\}$. $\{i3, i5\}$ is infrequent --> prune

L3:

$\{i1, i2, i3\}$
 $\{i1, i2, i5\}$.

C4:

$\{i1, i2, i3, i5\}$. $\{i1, i3, i5\}$ not frequent $\rightarrow$ prune